University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2021

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Section 1. Project Management [20 Marks]

Question 1 [13 Marks]

a. Consider the following five principles of Agile methods: [10]

A. Customer involvement
B. Incremental delivery
C. People not process
D. Embrace change
E. Maintain simplicity

Pick the two you consider most important and explain what they are and give a sentence or two on why you think they are so important?

ANSWER: *Can pick any two! The following is the answer to "what they are", [3] marks for each of the two. And [4] for reasons given for why they are important.*

Customer involvement: Customers should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and to evaluate the iterations of the system

Incremental delivery: The software is developed in increments with the customer specifying the requirements to be included in each increment.

People not process: The skills of the development team should be recognized and exploited. Team members should be left to develop their own ways of working without prescriptive processes.

Embrace change: Expect the system requirements to change and so design the system to accommodate these changes

Maintain simplicity: Focus on simplicity in both the software being developed and in the development process. Wherever possible, actively work to eliminate complexity from the system.

REASONS: *Anything that makes sense and has at least 2 points made [4]. Only one point made [2]. The important thing is that there are logical reasons. No logic = 0. Can even say refuse to choose between the options since they all hang together and do not make sense separately [4]*

b. Characterize the differences between waterfall and agile methods in terms of time, cost, and scope. [3]

HARD

In waterfall – scope is fixed, while time and cost will increase if customer objectives are not met as originally allocated, this leads to projects going over budget and over time.

+1 for capturing waterfall triangle

In agile – scope is not fixed, white time and cost are. Scope is adjusted and prioritized so that the most important bits are completed first and additional features are only completed if time and budget are available.

+1 for capturing agile triangle

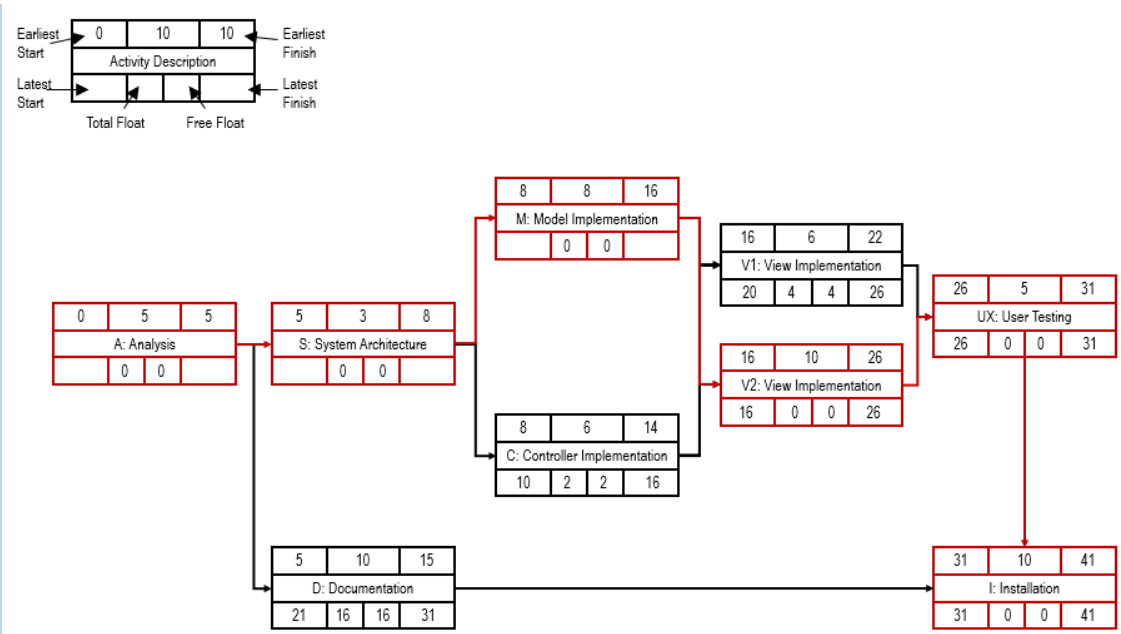
Differences – the main difference is in how scope is managed, governed a bit by availability of time and budget. For projects where the complete scope is prioritized and known from the beginning, this can only be achieved by increasing the budget/time where work is underestimated, where in agile development the focus is not on achieving a complete pre-specified project but instead on maximizing the customer needs to be met in a given amount of time.

+1 for coherent comparison (or characterization of differences) noting tradeoffs between waterfall and agile. *Simply giving the triangle for each but not comparing them directly (i.e. leaving differences implicity) does not earn this point.*

Question 2 [7 Marks]

a. Draw a network analysis diagram based on the following task list and identify the critical path. [7]

Task	Time	Dependencies
A: Analysis	5	
S: System Architecture Planning	3	A
M: Model Implementation	8	S
C: Controller Implementation	6	S
V1: View1 Implementation (User View)	6	M, C
V2: View2 Implementation (Admin View)	10	M, C
UX: User Testing	5	V1, V2
D: Documentation	10	A
I: Installation and Customer Hand Over	10	D, UX



[2] Dependencies: 2 points total, minus half point for each error

[2] Critical Path: 2 points for correct identification of critical path, 1 point for partially correct

Check the following numbers

[2] Earliest Finish: 1 point for getting I: Installation correct, ½ for V1: View, ½ for C: Controller

[1] Total float: Latest Start – Earliest Start (1 point for correct calculations, even if the numbers for LS and ES are incorrect)

[1] Free float: ES (next) – EF (current) (l point for correct calculations)

Section 2. Architecture [20 Marks]

Question 3 [6 Marks]

a. Explain cohesion and coupling, and what the goal of architectural design is in terms of those concepts. [3]

Coupling is the degree to which sub classes rely on base classes for their functionality. Classes that rely heavily on base classes for their functionally have high coupling+1, which is considered a design flaw. Cohesion refers to how the classes in an application work together coherently in order to provide the functionality of an application+1. Good applications aim for an architecture that provides low coupling and high cohesion+1

b. Define and discuss the Single Responsibility Principle in terms of cohesion and coupling. [2]

Making use of classes that have a single responsibility over a single functionality in an application ultimately leads to low coupling because the sub classes are not reliant on one god object that introduces a single point of failure or tight coupling.+1 The classes also provide high cohesion as all the classes with a single responsibility function together to provide the overall functionality of the application+1.

c. Compilers are an example of which kind of general application architecture? [1]

Language Processing Application Architecture =2

Question 4 [4 Marks]

a. A client on your current project has requested a new feature, explain what type of software maintenance this change request falls under. [2]

Perfective Maintenance +2

b. Briefly describe the maintenance process that would take place for the above-mentioned request. [2]

Perfective maintenance takes place when a new feature is requested via a change request. A business committee will meat and decide on the change impact. The new feature will be developed if the impact analysis deems it fit. +2 (Award one mark if the student is able to state that a new feature will be developed after a change request, but does not mention any further detail regarding the change request)

Question 5 [10 Marks]

Consider the Canzibe/Lwandile Network from our first week's tutorial.



In this tele-consultation, nurses in 3-5 remote clinics are connected to doctors in the hospital via long-distance wireless links. These links are not always available, and may disconnect due to power outages or link loss. The doctors and nurses communicate both asynchronously (store-and-forward messages) and via real-time video calls. Patient records can be attached to messages. If one clinic is disconnected from the hospital due to link outages, the other clinics must still work, provided availability of a connection to the hospital.

Key non-functional requirements include:

- Security and privacy of the data
- Reliability (no lost messages)
- Best-effort availability (subject to link outages)

a. Briefly explain what the most sensible underlying architectural design pattern to use is, and why you chose it. [6]

0	*No answer*
1	*Mentions an architecture but does not provide any motivations as to why the architecture was chosen*
2	*Mentions an architecture and provides a clear definition of the architecture and how this relates to at least one of the non-functional requirements.*
3	*Mentions an architecture not listed below (e.g. client/server without redundancy) with clear and accurate explanations that meet at least 2 of the non-functional requirements.*

4	***SOA or Microservices*** *Gives reasons and adequate description of how to implement this case as an SOA or Microservices architecture but is not explicit about how the architecture meets non-functional requirements* *OR* ***Client-server with redundancy*** *Gives reasons and adequate description of how the architecture fits the case, but is not explicit about how the architecture meets non-functional requirements*
5	***SOA or Microservices*** *Gives reasons and adequate description of how the architecture fits the case answer should be clearly justified, mentioning specific elements that link directly to specified non-func req (sec/priv, reliability, best-effort availability)* *OR* ***Client-server with redundancy*** *Gives reasons and adequate description of how the architecture fits the case. This is a more traditional approach – answer should be clearly justified, mentioning specific elements that link directly to specified non-func req (sec/priv, reliability, best-effort availability).*
6	***SOA or Microservices*** *Gives a well-thought-out explanation that clearly mentions the elements of the architecture that allows for privacy, reliability and availability is provided Separated out services that function independently of each other in SOA, separate APIs that function independently of each other in Micro Services or fail over cluster or any other redundancy technique in the client server / layered architecture)* ***Specifically mentions that if one service fails, others will continue running, which improves reliability and best-effort availaibility***

b. Use a UML diagram to illustrate the actual system showing all the major components/modules. [4]

The following is clearly represented in the UML diagram :

1. *The data store [1]*
2. *Redundant services/ API's/ Servers [2]*
3. *The interconnection of the clinics and hospital [1]*
4. *Repository of messages that could not be sent [1]*

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